

# MEDIBUS for Dräger Intensive Care Devices

Instructions for Use

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# For Your Safety and that of Your Patients

For correct and effective use of the apparatus and to avoid hazards it is essential to read the following recommendations and to act accordingly:

## **Strictly follow the Instructions for Use**

Any use of the apparatus requires full understanding and strict observation of these instructions. The apparatus is only to be used for purposes specified here.

## **Liability for proper function or damage**

The liability for the proper function of the software protocol is irrevocably transferred to the owner or operator if the software protocol is used in a manner not conforming to its intended use.

Dräger cannot be held responsible for damage caused by non-compliance with the recommendations given above. The warranty and liability provisions of the terms of sale and delivery of Dräger are likewise not modified by the recommendations given above.

Dräger Medizintechnik GmbH

# Intended Use

MEDIBUS is a software protocol intended to be used by two medical devices for exchanging data and control functions via their RS 232 interfaces.

This part of manual describes device dependent supported commands and data sets, port hardware and configurations for Dräger Intensive Care Devices. For a general description of the protocol please refer to the Instructions for Use "Dräger RS 232 MEDIBUS Protocol Definition" (order-no. 90 28 258).

**Any data transmitted via the MEDIBUS interface are intended only for information purposes and should not be used to derive therapeutical decisions.**

## Data Formats

The underscore character used in the format column in lists of measured data and alarm limits is transmitted as an ASCII "space" character (20H).

A '\*' ahead the format indicates that the value may be negative. In that case a '-' character will appear at the first space of the respective format.

Be aware that in a small number of cases the used format for a certain item may be different!

## Languages

Alarm phrases and text messages are given in

6 languages:

D german

GB english

F french

I italien

NL dutch

E spanish

## Alarm Phrases

Some alarm phrases contain abbreviations as follows:

| ASCII Short Form | Meaning |
|------------------|---------|
| \$&              | LOW     |
| "#               | HIGH    |
| '@               | ALARM   |

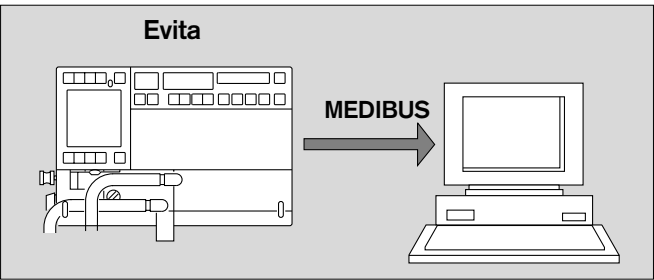
## Alarm Priorities

The alarm source is responsible for the priority.

The priority may be variable even if from the same source. (E.g. the priority may increase the longer an alarm is pending.)

# MEDIBUS Specification for Evita

## Device Connection



## Port Specification

|                    |                                                            |
|--------------------|------------------------------------------------------------|
| Connector          |                                                            |
| Type               | RS-232-C                                                   |
| Pins               | 9 pin Sub D (female)                                       |
| Pins               | 1 Housing                                                  |
| Pins               | 2 RXD                                                      |
| Pins               | 3 TXD                                                      |
| Pins               | 5 GND                                                      |
| Galvanic Isolation | 1,5 kV                                                     |
| Location           | rear side of Evita on optional extension board "EvitaLink" |
|                    | Label: RS 232                                              |

## Port Configuration

|           |            |
|-----------|------------|
| Channel A |            |
| Baudrate  | 19200 Baud |
| Databits  | 8          |
| Startbits | 1          |
| Stopbits  | 1          |
| Parity    | even       |

## Port Configuration

|           |                        |
|-----------|------------------------|
| Channel B |                        |
| Baudrate  | 1200, 2400, 4800, 9600 |
| Baudrate  | 19200 Baud             |
| Databits  | 8                      |
| Startbits | 1                      |
| Stopbits  | 1, 2                   |
| Parity    | none, odd, even        |

## Device Identification

|                 |                                           |
|-----------------|-------------------------------------------|
| ID-Number       | 8210                                      |
| Name            | 'Evita'                                   |
| MEDIBUS-Version | 03.00 for Device Version 01.00 and higher |

# MEDIBUS Specification for Evita

## Available Data

MEDIBUS is available only with optional hardware extension "EvitaLink"\*.

Current Measured Data, Low and High Alarm Limits, Alarmstatus, Device Settings, Text Messages and Realtime-Data are available since Evita device version 13.03/14.03 and EvitaLink device version 2.01.

## Commands

### Transmitted Commands

| Code | Command-Specification          |
|------|--------------------------------|
| 30H  | Do nothing (NOP)               |
| 51H  | Initialize Communication (ICC) |
| 52H  | Request Device Identification  |

### Processed and responded Commands

| Code | Command-Specification             |
|------|-----------------------------------|
| 24H  | Request current DATA              |
| 25H  | Request current LOW ALARM LIMITS  |
| 26H  | Request current HIGH ALARM LIMITS |
| 27H  | Request current ALARMS            |
| 29H  | Request current DEVICE SETTINGS   |
| 2AH  | Request current TEXT MESSAGES     |
| 30H  | Do nothing (NOP)                  |
| 4AH  | Configure Data Response           |
| 51H  | Initialize Communication (ICC)    |
| 52H  | Request Device Identification     |
| 53H  | Request Realtime Configuration    |
| 54H  | Configure Realtime Transmission   |
| 55H  | Stop Communication                |
| 6AH  | Device Specific                   |

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\* RS 232 Communication Interface for Evita and Evita 2

# MEDIBUS Specification for Evita

## Measured Data, Low and High Alarm Limits

Airway related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description               | Unit     | Format | M | LL | HL |
|------|--------------------------------|----------|--------|---|----|----|
| 07H  | Compliance L/bar               | L/bar    | _XXX   | x |    |    |
| 08H  | Resistance                     | mbar/L/s | XXX_   | x |    |    |
| 71H  | Minimal Airway Pressure        | mbar     | *_XX_  | x |    |    |
| 72H  | Occlusion Pressure             | mbar     | XX.X   | x |    |    |
| 73H  | Mean Breathing Press.          | mbar     | *_XX_  | x |    |    |
| 74H  | Plateau Pressure               | mbar     | *_XX_  | x |    |    |
| 78H  | PEEP Breathing Press.          | mbar     | *_XX_  | x |    |    |
| 79H  | Intrinsic PEEP Breath. Press.  | mbar     | XX.X   | x |    |    |
| 7DH  | Peak Breathing Press.          | mbar     | *_XX_  | x |    | x  |
| 81H  | Trapped Volume                 | mL       | _XXX   | x |    |    |
| 82H  | Tidal Volume L                 | L        | X.XX   | x |    |    |
| B5H  | Spontaneous Respiratory Rate   | 1/min    | XXX_   | x |    |    |
| B7H  | Spontaneous Minute Volume      | L/min    | XX.X   | x |    |    |
| B9H  | Respiratory Minute Volume      | L/min    | XX.X   | x | x  | x  |
| C1H  | Airway Temperature             | °C       | _XX_   | x |    |    |
| D6H  | Resp. Rate (Vol./Flow)(Pediat) | 1/min    | XXX_   | x |    | x  |

A '\*' ahead the format indicates that the value may be negative. In that case a '-' character precedes the numeric value.

## O<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description     | Unit | Format | M | LL | HL |
|------|----------------------|------|--------|---|----|----|
| F0H  | Insp. O <sub>2</sub> | %    | XXX_   | x |    |    |

# MEDIBUS Specification for Evita

## Realtime Data

| Code | Realtime-Data                  | Unit  |
|------|--------------------------------|-------|
| 00H  | Airway Pressure                | mbar  |
| 01H  | Flow (insp./exp.)              | L/min |
| 03H  | Resp. Volume since insp. Begin | mL    |
| 24H  | Expiratory Volume              | mL    |

## Realtime Sync-Commands

Realtime Sync-Commands will be sent when Realtime Data are enabled.

| Command-Code | Argument | Command-Specification                    |
|--------------|----------|------------------------------------------|
| C6H          | C0H      | Start of Ventilator<br>Inspiratory Cycle |

# MEDIBUS Specification for Evita

## Alarm Messages

Alarm phrases are sent in english only.

Note: Some alarm priorities are configurable by user!

### Airway related Alarms

| Code | Alarm Description                         | Priority | Alarm phrase |
|------|-------------------------------------------|----------|--------------|
| 10H  | Airway Pressure > high Limit              | 27       | PAW HIGH     |
| 19H  | Minute Volume < low Limit                 | 26       | MIN VOL LOW  |
| 33H  | Volume not constant                       | 26/3     | VOL INCONST  |
| 40H  | Tachypnea Alarm disabled                  | 1        | TACHYPN'@OFF |
| 5EH  | Volume Alarm disabled                     | 1        | VOL ALRM OFF |
| 90H  | Respiratory Rate > high Limit             | 26       | RESP RATE HI |
| 98H  | Apnoe detected by Evita                   | 27/8     | APNEA EVITA  |
| 9BH  | Minute Volume > high Limit                | 26       | MIN VOL HIGH |
| A4H  | Volume Measurement inoperable (Alarm)     | 27/8     | VOL ERR      |
| ACH  | MIN VOL Alarm disabled                    | 1        | MV ALRM OFF  |
| ADH  | Pressure Measurement inoperable           | 27       | PRESS ERR    |
| B8H  | Airway Temperature Measurement inoperable | 25       | AW-TEMP INOP |
| B9H  | Check Airway Temperature Sensor           | 25       | AW-TEMP SENS |
| BAH  | Airway Temperature > high Limit           | 25       | AW-TEMP HIGH |

### Miscellaneous Alarms

| Code | Alarm Description              | Priority | Alarm phrase  |
|------|--------------------------------|----------|---------------|
| 78H  | Communication Error RS232 Port | 1        | RS232 COM ERR |
| 7CH  | Internal Communication Error   | 1        | INT COM ERR   |

# MEDIBUS Specification for Evita

## O<sub>2</sub> related Alarms

| Code | Alarm Description                                  | Priority | Alarm phrase          |
|------|----------------------------------------------------|----------|-----------------------|
| 08H  | Insp. Oxygen < low Limit (Alarm)                   | 26       | % O <sub>2</sub> LOW  |
| 37H  | % Oxygen > high Limit (Caution)                    | 23       | % O <sub>2</sub> HIGH |
| A1H  | Insp O <sub>2</sub> Measurement inoperable (Alarm) | 28       | % O <sub>2</sub> ERR  |
| BEH  | O <sub>2</sub> Measurement inoperable (Advisory)   | 8        | % O <sub>2</sub> ERR  |
| BFH  | Insp O <sub>2</sub> > high Limit (Advisory)        | 7        | % O <sub>2</sub> HIGH |
| C0H  | Insp O <sub>2</sub> < low Limit (Advisory)         | 7        | % O <sub>2</sub> LOW  |

## Ventilator related Alarms

| Code | Alarm Description                        | Priority | Alarm phrase             |
|------|------------------------------------------|----------|--------------------------|
| 12H  | Check Air Supply                         | 31/2     | AIR SUPPLY ?             |
| 3AH  | Assisted Spontaneous Breathing > 4 sec.  | 26       | ASB > 4 SEC              |
| 9AH  | Disconnection Ventilator                 | 27       | PAW LOW                  |
| 9FH  | Problems with Respirator (Evita)         | 25       | EVITA ERR                |
| B0H  | Check Expiration-Valve                   | 27       | EXP-VALVE ?              |
| B7H  | Too high Respirator Device Temp. (Alarm) | 25       | COOLING INOP             |
| BBH  | Sigh Mode active                         | 2        | SIGH ON                  |
| BCH  | Breathing System vented                  | 10       | SYSTEM OPEN              |
| BDH  | Check O <sub>2</sub> Supply (Advisory)   | 31/10    | LO O <sub>2</sub> SUPPLY |
| C2H  | Gas Mixer inoperable (Advisory)          | 8        | MIXER INOP               |
| C3H  | Time limited Respiratory Volume          | 7        | TIME LIMITED             |
| C4H  | Pressure limited Respiratory Volume      | 2        | PRESSURE LTD             |
| C5H  | High Respirator Device Temp. (Advisory)  | 10       | COOLING INOP             |
| C6H  | Respirator Synchronisation inoperable    | 26       | SYNCHRO INOP             |
| C7H  | Fail to Cycle                            | 27       | CYCLE FAILED             |
| C8H  | Gas Mixer inoperable (Alarm)             | 29       | MIXER INOP               |

## Device Settings

| Code | Setting Description        | Unit  | Format |
|------|----------------------------|-------|--------|
| 01H  | Insp. Oxygen               | %     | _XXX_  |
| 02H  | Max. insp. Flow            | L/min | XXX.X  |
| 04H  | Insp. Tidal Volume         | L     | X.XXX  |
| 07H  | I : E I-Part               | -     | _XX.X  |
| 08H  | I : E E-Part               | -     | XXX.X  |
| 09H  | Frequency IMV (SIMV)       | 1/min | XXX.X  |
| 0AH  | Frequency IPPV             | 1/min | XXX.X  |
| 0BH  | PEEP (CPAP)                | mbar  | _XX.X  |
| 0CH  | Intermittend PEEP          | mbar  | __XX_  |
| 0DH  | BIPAP low Pressure         | mbar  | __XX_  |
| 0EH  | BIPAP high Pressure        | mbar  | __XX_  |
| 0FH  | BIPAP low Time             | sec   | _XX.X  |
| 10H  | BIPAP high Time            | sec   | _XX.X  |
| 11H  | Apnea Time                 | sec   | __XX_  |
| 12H  | Assisted spon. Breath.     | mbar  | __XX_  |
| 13H  | Max. insp. Airway Pressure | mbar  | XXX.X  |
| 15H  | Trigger Pressure           | mbar  | _XX.X  |
| 16H  | Tachyapnea Frequency       | 1/min | _XXX_  |
| 17H  | Tachyapnea Duration        | sec   | _XXX_  |
| 29H  | Flow Trigger               | L/min | __XX_  |
| 2EH  | ASB Ramp                   | sec   | __XX_  |

# MEDIBUS Specification for Evita

## Text Messages

Text Messages are sent in english only.

| Code | Text Message           |
|------|------------------------|
| 01H  | Mode IPPV              |
| 02H  | Mode IPPV/ASSIST       |
| 04H  | Mode CPPV              |
| 05H  | Mode CPPV/ASSIST       |
| 06H  | Mode SIMV              |
| 07H  | Mode SIMV/ASB          |
| 08H  | Mode SB                |
| 09H  | Mode ASB               |
| 0AH  | Mode CPAP              |
| 0BH  | Mode CPAP/ASB          |
| 0CH  | Mode MMV               |
| 0DH  | Mode MMV/ASB           |
| 0EH  | Mode BIPAP             |
| 0FH  | Mode SYNCHRON MASTER   |
| 10H  | Mode SYNCHRON SLAVE    |
| 11H  | Mode APNEA VENTILATION |
| 12H  | Mode DS                |
| 18H  | Mode BIPAP-SIMV        |
| 19H  | Mode BIPAP-SIMV/ASB    |
| 1AH  | Mode BIPAP-APRV        |

# MEDIBUS Specification for Evita

## Device specific Commands

Evita uses the device specific command (code 6AH) to transmit some additional information which is not part of the MEDIBUS standard. The device specific command has the following format:

| ESC    | Command Code (6AH) | Argument | Checksum | CR |
|--------|--------------------|----------|----------|----|
| 0      | 1                  | 2        | 3        | 5  |
| 6 Byte |                    |          |          |    |

The 'Argument' is a one byte ASCII character specifying the device specific command:

| Argument | Meaning                   |
|----------|---------------------------|
| '2'      | Request Key Status        |
| '3'      | Request EvitaLink Version |
| '5'      | Request EvitaLink Channel |

## Device specific Responses

### Key Status

This response returns all active keys on a 'Request Key Status' command in the following format:

| SOH       | Command Echo (6AH) | Argument Echo ('2') | Key Code 1 | ... | Key Code m | Checksum | CR   |
|-----------|--------------------|---------------------|------------|-----|------------|----------|------|
| 0         | 1                  | 2                   | 3          | 5   | 2m+1       | 2m+3     | 2m+4 |
| 2m+5 Byte |                    |                     |            |     |            |          |      |

The Key Code is a two byte ASCII field holding an identifier for each currently active key. The meaning of Key Code is given in the following table:

| Key Code | Meaning                   |
|----------|---------------------------|
| '32'     | Alarm Silence active      |
| '33'     | Nebulizer active          |
| '34'     | Display stop on           |
| '35'     | Oxygen Calibration active |
| '36'     | Oxygen Monitoring off     |
| '37'     | Suction active            |
| '38'     | Flow Calibration active   |

# MEDIBUS Specification for Evita

## EvitaLink Version

This command returns the software version of EvitaLink in the following format:

| SOH     | Command Echo (6AH) | Argument Echo ('3') | Version | Checksum | CR |
|---------|--------------------|---------------------|---------|----------|----|
| 0       | 1                  | 2                   | 3       | 8        | 9  |
| 10 Byte |                    |                     |         |          |    |

The Version is a five byte ASCII field.  
Example: Version 2.00 = '02.00'

## EvitaLink Channel

This command returns the number of the EvitaLink channel as a one byte ASCII Code:

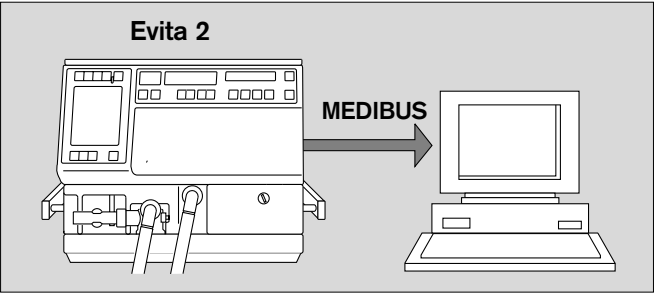
| Channel | ASCII Code |
|---------|------------|
| A       | '0'        |
| B       | '1'        |

The response format is specified as follows:

| SOH    | Command Echo (6AH) | Argument Echo ('5') | Channel | Checksum | CR |
|--------|--------------------|---------------------|---------|----------|----|
| 0      | 1                  | 2                   | 3       | 4        | 5  |
| 6 Byte |                    |                     |         |          |    |

# MEDIBUS Specification for Evita 2

## Device Connection



## Port Specification

|                    |                                                              |
|--------------------|--------------------------------------------------------------|
| Connector          |                                                              |
| Type               | RS-232-C                                                     |
| Pins               | 9 pin Sub D (female)                                         |
| Pins               | 1 Housing                                                    |
| Pins               | 2 RXD                                                        |
| Pins               | 3 TXD                                                        |
| Pins               | 5 GND                                                        |
| Galvanic Isolation | 1,5 kV                                                       |
| Location           | rear side of Evita 2 on optional extension board "EvitaLink" |
|                    | Label: RS 232                                                |

## Port Configuration

|           |            |
|-----------|------------|
| Channel A |            |
| Baudrate  | 19200 Baud |
| Databits  | 8          |
| Startbits | 1          |
| Stopbits  | 1          |
| Parity    | even       |

## Port Configuration

|           |                        |
|-----------|------------------------|
| Channel B |                        |
| Baudrate  | 1200, 2400, 4800, 9600 |
| Baudrate  | 19200 Baud             |
| Databits  | 8                      |
| Startbits | 1                      |
| Stopbits  | 1, 2                   |
| Parity    | none, odd, even        |

## Device Identification

|                 |                                           |
|-----------------|-------------------------------------------|
| ID-Number       | 8200                                      |
| Name            | 'Evita 2'                                 |
| MEDIBUS-Version | 03.00 for Device Version 01.00 and higher |

# MEDIBUS Specification for Evita 2

## Available Data

MEDIBUS is available only with optional hardware extension "EvitaLink"\*.

Current Measured Data, Low and High Alarm Limits, Alarmstatus, Device Settings, Text Messages and Realtime-Data are available since Evita 2 device version 1.00 and EvitaLink device version 1.00 and EvitaLink device version 2.01.

## Commands

### Transmitted Commands

| Code | Command-Specification          |
|------|--------------------------------|
| 30H  | Do nothing (NOP)               |
| 51H  | Initialize Communication (ICC) |
| 52H  | Request Device Identification  |

### Processed and responded Commands

| Code | Command-Specification             |
|------|-----------------------------------|
| 24H  | Request current DATA              |
| 25H  | Request current LOW ALARM LIMITS  |
| 26H  | Request current HIGH ALARM LIMITS |
| 27H  | Request current ALARMS            |
| 29H  | Request current DEVICE SETTINGS   |
| 2AH  | Request current TEXT MESSAGES     |
| 30H  | Do nothing (NOP)                  |
| 4AH  | Configure Data Response           |
| 51H  | Initialize Communication (ICC)    |
| 52H  | Request Device Identification     |
| 53H  | Request Realtime Configuration    |
| 54H  | Configure Realtime Transmission   |
| 55H  | Stop Communication                |
| 6AH  | Device Specific                   |

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\* RS 232 Communication Interface for Evita and Evita 2

## Measured Data, Low and High Alarm Limits

Airway related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description               | Unit     | Format | M | LL | HL |
|------|--------------------------------|----------|--------|---|----|----|
| 07H  | Compliance L/bar               | L/bar    | _XXX   | x |    |    |
| 08H  | Resistance                     | mbar/L/s | XXX_   | x |    |    |
| 71H  | Minimal Airway Pressure        | mbar     | *_XX_  | x |    |    |
| 72H  | Occlusion Pressure             | mbar     | XX.X   | x |    |    |
| 73H  | Mean Breathing Press.          | mbar     | *_XX_  | x |    |    |
| 74H  | Plateau Pressure               | mbar     | *_XX_  | x |    |    |
| 78H  | PEEP Breathing Press.          | mbar     | *_XX_  | x |    |    |
| 79H  | Intrinsic PEEP Breath. Press.  | mbar     | XX.X   | x |    |    |
| 7DH  | Peak Breathing Press.          | mbar     | *_XX_  | x |    | x  |
| 81H  | Trapped Volume                 | mL       | _XXX   | x |    |    |
| 82H  | Tidal Volume L                 | L        | X.XX   | x |    |    |
| B5H  | Spontaneous Respiratory Rate   | 1/min    | XXX_   | x |    |    |
| B7H  | Spontaneous Minute Volume      | L/min    | XX.X   | x |    |    |
| B9H  | Respiratory Minute Volume      | L/min    | XX.X   | x | x  | x  |
| C1H  | Airway Temperature             | °C       | _XX_   | x |    |    |
| D6H  | Resp. Rate (Vol./Flow)(Pediat) | 1/min    | XXX_   | x |    | x  |

A '\*' ahead the format indicates that the value may be negative. In that case a '-' character precedes the numeric value.

## CO<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits<sup>1)</sup>

| Code | Data-Description                 | Unit   | Format | M | LL | HL |
|------|----------------------------------|--------|--------|---|----|----|
| 09H  | CO <sub>2</sub> -Production      | mL/min | XXX_   | x |    |    |
| 89H  | Dead Space (Vds)                 | mL     | _XXX   | x |    |    |
| 8AH  | Relative Dead Space              | %      | _XX_   | x |    |    |
| DBH  | Endtidal CO <sub>2</sub> in %    | %      | XX.X   | x | x  | x  |
| E3H  | Endtidal CO <sub>2</sub> in kPa  | kPa    | XX.X   | x | x  | x  |
| E6H  | Endtidal CO <sub>2</sub> in mmHg | mmHg   | _XX_   | x | x  | x  |

1) Available with CO<sub>2</sub> measurement option.

# MEDIBUS Specification for Evita 2

## O<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description     | Unit | Format | M | LL | HL |
|------|----------------------|------|--------|---|----|----|
| F0H  | Insp. O <sub>2</sub> | %    | XXX_   | x |    |    |

## Realtime Data

| Code              | Realtime-Data                  | Unit  |
|-------------------|--------------------------------|-------|
| 00H               | Airway Pressure                | mbar  |
| 01H               | Flow (insp./exp.)              | L/min |
| 03H               | Resp. Volume since insp. Begin | mL    |
| 06H <sup>1)</sup> | Exp. CO <sub>2</sub>           | mmHg  |
| 07H <sup>1)</sup> | Exp. CO <sub>2</sub>           | kPa   |
| 08H <sup>1)</sup> | Exp. CO <sub>2</sub>           | %     |
| 24H <sup>1)</sup> | Expiratory Volume              | mL    |

## Realtime Sync-Commands

Realtime Sync-Commands will be sent when Realtime Data are enabled.

| Command-Code | Argument | Command-Specification                 |
|--------------|----------|---------------------------------------|
| C6H          | C0H      | Start of Ventilator Inspiratory Cycle |

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1) Available since EvitaLink Software 2.00

## Alarm Messages

Alarm phrases are sent in english only.

Note: Some alarm priorities are configurable by user!

### Airway related Alarms

| Code              | Alarm Description                         | Priority | Alarm phrase |
|-------------------|-------------------------------------------|----------|--------------|
| 10H               | Airway Pressure > high Limit              | 27       | PAW HIGH     |
| 19H               | Minute Volume < low Limit                 | 26       | MIN VOL LOW  |
| 33H               | Volume not constant                       | 26/3     | VOL INCONST  |
| 40H               | Tachypnea Alarm disabled                  | 1        | TACHYPN'@OFF |
| 42H <sup>1)</sup> | Check Flow Sensor                         | 27       | FLOW SENSOR? |
| 5EH               | Volume Alarm disabled                     | 1        | VOL ALRM OFF |
| 90H               | Respiratory Rate > high Limit             | 26       | RESP RATE HI |
| 98H               | Apnoe detected by Evita                   | 27       | APNEA EVITA  |
| 9BH               | Minute Volume > high Limit                | 26       | MIN VOL HIGH |
| A4H               | Volume Measurement inoperable (Alarm)     | 27/8     | VOL ERR      |
| ACH               | MIN VOL Alarm disabled                    | 1        | MV ALRM OFF  |
| ADH               | Pressure Measurement inoperable           | 27       | PRESS ERR    |
| B8H               | Airway Temperature Measurement inoperable | 25       | AW-TEMP INOP |
| B9H               | Check Airway Temperature Sensor           | 25       | AW-TEMP SENS |
| BAH               | Airway Temperature > high Limit           | 25       | AW-TEMP HIGH |
| DAH <sup>1)</sup> | PEEP high Pressure Limit                  | 27       | PEEP HIGH    |

### CO<sub>2</sub> related Alarms<sup>2)</sup>

| Code | Alarm Description                               | Priority | Alarm phrase |
|------|-------------------------------------------------|----------|--------------|
| 27H  | Endtidal CO <sub>2</sub> < low Limit            | 25       | ET CO2 LOW   |
| 28H  | Endtidal CO <sub>2</sub> > high Limit           | 25       | ET CO2 HIGH  |
| 3EH  | CO <sub>2</sub> not calibrated                  | 25/7     | CO2 NOT CAL  |
| 57H  | CO <sub>2</sub> Alarm disabled                  | 1        | CO2 ALRM OFF |
| 63H  | CO <sub>2</sub> Mon. in Low Acc. Mode (warm up) | 1        | CO2 WARM UP  |
| 64H  | CO <sub>2</sub> Window occluded                 | 25/7     | CLEAN CO2    |
| 6AH  | CO <sub>2</sub> Device Failure                  | 25/7     | CO2 ERR      |
| D9H  | CO <sub>2</sub> Sensor disconnected or fault    | 25/7     | CO2 SENS ?   |

1) Availabe since Evita 2 Software 2.00

2) Available with CO<sub>2</sub> measurement option

# MEDIBUS Specification for Evita 2

## Miscellaneous Alarms

| Code | Alarm Description              | Priority | Alarm phrase  |
|------|--------------------------------|----------|---------------|
| 78H  | Communication Error RS232 Port | 1        | RS232 COM ERR |
| 7CH  | Internal Communication Error   | 1        | INT COM ERR   |

## O<sub>2</sub> related Alarms

| Code | Alarm Description                                  | Priority | Alarm phrase          |
|------|----------------------------------------------------|----------|-----------------------|
| 08H  | Insp. Oxygen < low Limit (Alarm)                   | 26       | % O <sub>2</sub> LOW  |
| 37H  | % Oxygen > high Limit (Caution)                    | 23       | % O <sub>2</sub> HIGH |
| A1H  | Insp O <sub>2</sub> Measurement inoperable (Alarm) | 28       | % O <sub>2</sub> ERR  |
| BEH  | O <sub>2</sub> Measurement inoperable (Advisory)   | 8        | % O <sub>2</sub> ERR  |
| BFH  | Insp O <sub>2</sub> > high Limit (Advisory)        | 7        | % O <sub>2</sub> HIGH |
| C0H  | Insp O <sub>2</sub> < low Limit (Advisory)         | 7        | % O <sub>2</sub> LOW  |

## Ventilator related Alarms

| Code | Alarm Description                        | Priority | Alarm phrase             |
|------|------------------------------------------|----------|--------------------------|
| 12H  | Check Air Supply                         | 31/2     | AIR SUPPLY ?             |
| 3AH  | Assisted Spontaneous Breathing > 4 sec.  | 26       | ASB > 4 SEC              |
| 9AH  | Disconnection Ventilator                 | 27       | PAW LOW                  |
| 9FH  | Problems with Respirator (Evita)         | 25       | EVITA ERR                |
| B0H  | Check Expiration-Valve                   | 27       | EXP-VALVE ?              |
| B7H  | Too high Respirator Device Temp. (Alarm) | 25       | COOLING INOP             |
| BBH  | Sigh Mode active                         | 2        | SIGH ON                  |
| BCH  | Breathing System vented                  | 10       | SYSTEM OPEN              |
| BDH  | Check O <sub>2</sub> Supply (Advisory)   | 31/10    | LO O <sub>2</sub> SUPPLY |
| C2H  | Gas Mixer inoperable (Advisory)          | 8        | MIXER INOP               |
| C3H  | Time limited Respiratory Volume          | 7        | TIME LIMITED             |
| C4H  | Pressure limited Respiratory Volume      | 2        | PRESSURE LTD             |
| C5H  | High Respirator Device Temp. (Advisory)  | 10       | COOLING INOP             |
| C6H  | Respirator Synchronisation inoperable    | 26       | SYNCHRO INOP             |
| C7H  | Fail to Cycle                            | 27       | CYCLE FAILED             |
| C8H  | Gas Mixer inoperable (Alarm)             | 29       | MIXER INOP               |

## Device Settings

| Code              | Setting Description        | Unit  | Format |
|-------------------|----------------------------|-------|--------|
| 01H               | Insp. Oxygen               | %     | _XXX_  |
| 02H               | Max. insp. Flow            | L/min | XXX.X  |
| 04H               | Insp. Tidal Volume         | L     | X.XXX  |
| 07H               | I : E I-Part               | -     | _XX.X  |
| 08H               | I : E E-Part               | -     | XXX.X  |
| 09H               | Frequency IMV (SIMV)       | 1/min | XXX.X  |
| 0AH               | Frequency IPPV             | 1/min | XXX.X  |
| 0BH               | PEEP (CPAP)                | mbar  | _XX.X  |
| 0CH               | Intermittend PEEP          | mbar  | __XX_  |
| 0DH               | BIPAP low Pressure         | mbar  | __XX_  |
| 0EH               | BIPAP high Pressure        | mbar  | __XX_  |
| 0FH               | BIPAP low Time             | sec   | _XX.X  |
| 10H               | BIPAP high Time            | sec   | _XX.X  |
| 11H               | Apnea Time                 | sec   | __XX_  |
| 12H               | Assisted spon. Breath.     | mbar  | __XX_  |
| 13H               | Max. insp. Airway Pressure | mbar  | XXX.X  |
| 15H               | Trigger Pressure           | mbar  | _XX.X  |
| 16H               | Tachyapnea Frequency       | 1/min | _XXX_  |
| 17H               | Tachyapnea Duration        | sec   | _XXX_  |
| 29H               | Flow Trigger               | L/min | __XX_  |
| 2EH <sup>1)</sup> | ASB Ramp                   | sec   | __XX_  |

1) Available since Evita 2 software 2.00

# MEDIBUS Specification for Evita 2

## Text Messages

Text Messages are sent in english only.

| Code                 | Text Message           |
|----------------------|------------------------|
| 01H                  | Mode IPPV              |
| 02H                  | Mode IPPV/ASSIST       |
| 04H                  | Mode CPPV              |
| 05H                  | Mode CPPV/ASSIST       |
| 06H                  | Mode SIMV              |
| 07H                  | Mode SIMV/ASB          |
| 08H                  | Mode SB                |
| 09H                  | Mode ASB               |
| 0AH                  | Mode CPAP              |
| 0BH                  | Mode CPAP/ASB          |
| 0CH                  | Mode MMV               |
| 0DH                  | Mode MMV/ASB           |
| 0EH                  | Mode BIPAP             |
| 0FH                  | Mode SYNCHRON MASTER   |
| 10H                  | Mode SYNCHRON SLAVE    |
| 11H                  | Mode APNEA VENTILATION |
| 12H                  | Mode DS                |
| 18H                  | Mode BIPAP-SIMV        |
| 19H                  | Mode BIPAP-SIMV/ASB    |
| 1AH                  | Mode BIPAP-APRV        |
| 20H <sup>1)</sup>    | Mode Adults            |
| 21H <sup>1)</sup>    | Mode Neonates          |
| 22H <sup>1) 2)</sup> | mmHg                   |
| 23H <sup>1) 2)</sup> | kPa                    |
| 24H <sup>1) 2)</sup> | %                      |

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1) Availabe since EvitaLink Software 2.01

2) Selected CO<sub>2</sub> unit

# MEDIBUS Specification for Evita 2

## Device specific Commands

Evita/Evita 2 uses the device specific command (code 6AH) to transmit some additional information which is not part of the MEDIBUS standard. The device specific command has the following format:

| ESC    | Command Code (6AH) | Argument | Checksum | CR |
|--------|--------------------|----------|----------|----|
| 0      | 1                  | 2        | 3        | 5  |
| 6 Byte |                    |          |          |    |

The 'Argument' is a one byte ASCII character specifying the device specific command:

| Argument | Meaning                   |
|----------|---------------------------|
| '2'      | Request Key Status        |
| '3'      | Request EvitaLink Version |
| '4'      | Request Patient Type      |
| '5'      | Request EvitaLink Channel |

## Device specific Responses

### Key Status

This response returns all active keys on a 'Request Key Status' command in the following format:

| SOH       | Command Echo (6AH) | Argument Echo ('2') | Key Code 1 | ... | Key Code m | Checksum | CR   |
|-----------|--------------------|---------------------|------------|-----|------------|----------|------|
| 0         | 1                  | 2                   | 3          | 5   | 2m+1       | 2m+3     | 2m+4 |
| 2m+5 Byte |                    |                     |            |     |            |          |      |

The Key Code is a two byte ASCII field holding an identifier for each currently active key. The meaning of Key Code is given in the following table:

| Key Code | Meaning                   |
|----------|---------------------------|
| '32'     | Alarm Silence active      |
| '33'     | Nebulizer active          |
| '34'     | Display stop on           |
| '35'     | Oxygen Calibration active |
| '36'     | Oxygen Monitoring off     |
| '37'     | Suction active            |
| '38'     | Flow Calibration active   |

# MEDIBUS Specification for Evita 2

## EvitaLink Version

This response returns the software version of EvitaLink in the following format:

| SOH     | Command Echo (6AH) | Argument Echo ('3') | Version | Checksum | CR |
|---------|--------------------|---------------------|---------|----------|----|
| 0       | 1                  | 2                   | 3       | 8        | 9  |
| 10 Byte |                    |                     |         |          |    |

The Version is a five byte ASCII field.  
Example: Version 2.00 = '02.00'

## Patient Type

This response returns the type of patient which has been chosen in Evita 2 in the following format:

| SOH      | Command Echo (6AH) | Argument Echo ('4') | Patient | Checksum | CR  |
|----------|--------------------|---------------------|---------|----------|-----|
| 0        | 1                  | 2                   | 3       | m+3      | m+4 |
| m+5 Byte |                    |                     |         |          |     |

Patient is a ASCII byte field which is given in the table below:

| Patient    | ASCII Byte field | Field lenght |
|------------|------------------|--------------|
| Adult      | 'Adult'          | 5            |
| Paediatric | 'Paediatric'     | 9            |

## EvitaLink Channel

This response returns the number of the EvitaLink channel as a one byte ASCII Code:

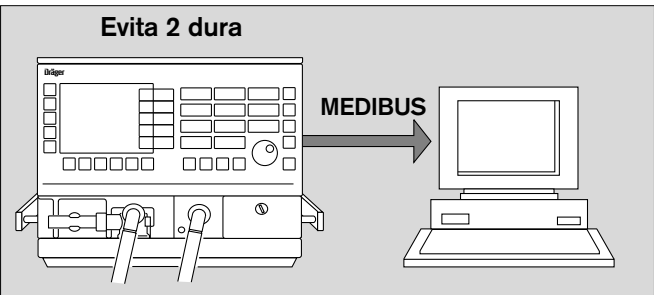
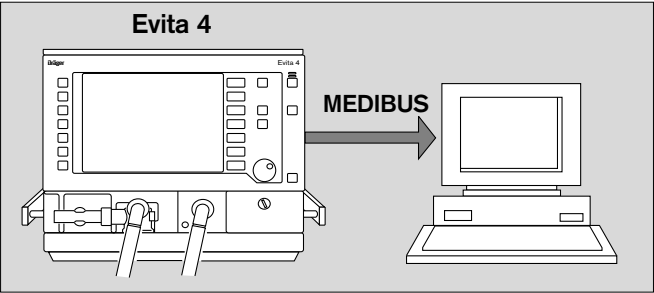
| Channel | ASCII Code |
|---------|------------|
| A       | '0'        |
| B       | '1'        |

The response format is specified as follows:

| SOH    | Command Echo (6AH) | Argument Echo ('5') | Channel | Checksum | CR |
|--------|--------------------|---------------------|---------|----------|----|
| 0      | 1                  | 2                   | 3       | 4        | 5  |
| 6 Byte |                    |                     |         |          |    |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

## Device Connection



## Port Specification

|                    |                                     |
|--------------------|-------------------------------------|
| Connector          |                                     |
| Type               | RS-232-C                            |
| Pins               | 9 pin Sub D (female)                |
| Pins               | 1 Housing                           |
| Pins               | 2 RXD                               |
| Pins               | 3 TXD                               |
| Pins               | 5 GND                               |
| Galvanic Isolation | 1,5 kV                              |
| Location           | rear side of Evita 4 / Evita 2 dura |
| Label              | COM1                                |
| Label              | or on optional extension board      |
| Label              | "Evita 4 Link", Label COM2 or       |
| Label              | COM3                                |

## Port Configuration

|           |                        |
|-----------|------------------------|
| Channel B |                        |
| Baudrate  | 1200, 2400, 4800, 9600 |
|           | 19200 Baud             |
| Databits  | 8                      |
| Startbits | 1                      |
| Stopbits  | 1, 2                   |
| Parity    | none, odd, even        |

## Device Identification for Evita 4

|                 |                                |
|-----------------|--------------------------------|
| ID-Number       | 8240                           |
| Name            | 'Evita 4'                      |
| MEDIBUS-Version | 04.00 for Device Version 01.00 |
|                 | and higher                     |

## Device Identification for Evita 2 dura

|                 |                                |
|-----------------|--------------------------------|
| ID-Number       | 8230                           |
| Name            | 'Evita 2 dura'                 |
| MEDIBUS-Version | 04.00 for Device Version 01.00 |
|                 | and higher                     |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

## Available Data

MEDIBUS is available since Evita 4 / Evita 2 dura device version 01.00 and higher.

All messages are transfered in english only independent of the language setting.

## Commands

### Transmitted Commands

| Code | Command-Specification          |
|------|--------------------------------|
| 30H  | Do nothing (NOP)               |
| 51H  | Initialize Communication (ICC) |
| 52H  | Request Device Identification  |

### Processed and responded Command

| Code | Command-Specification               |
|------|-------------------------------------|
| 24H  | Request current DATA                |
| 25H  | Request current LOW ALARM LIMITS    |
| 26H  | Request current HIGH ALARM LIMITS   |
| 27H  | Request current ALARMS (Codepage 1) |
| 29H  | Request current DEVICE SETTINGS     |
| 2AH  | Request current TEXT MESSAGES       |
| 2EH  | Request current ALARMS (Codepage 2) |
| 30H  | Do nothing (NOP)                    |
| 4AH  | Configure Data Response             |
| 51H  | Initialize Communication (ICC)      |
| 52H  | Request Device Identification       |
| 53H  | Request Realtime Configuration      |
| 54H  | Configure Realtime Transmission     |
| 55H  | Stop Communication                  |
| 6AH  | Device Specific                     |

## Measured Data, Low and High Alarm Limits

Airway related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code              | Data-Description          | Unit     | Format | M | LL              | HL              |
|-------------------|---------------------------|----------|--------|---|-----------------|-----------------|
| 06H <sup>1)</sup> | Compliance                | L/bar    | XX.X   | x |                 |                 |
| 07H <sup>1)</sup> | Compliance                | L/bar    | _XXX   | x |                 |                 |
| 0BH <sup>1)</sup> | Resistance                | mbar/L/s | XX.X   | x |                 |                 |
| 08H <sup>1)</sup> | Resistance                | mbar/L/s | XXX_   | x |                 |                 |
| 71H               | Minimum Airway Pressure   | mbar     | *XXX_  | x |                 |                 |
| 72H               | Occlusion Pressure        | mbar     | XX.X   | x |                 |                 |
| 73H               | Mean Airway Pressure      | mbar     | *XXX_  | x |                 |                 |
| 74H               | Plateau Airway Pressure   | mbar     | *XXX_  | x |                 |                 |
| 78H               | PEEP Airway Pressure      | mbar     | *XXX_  | x |                 |                 |
| 79H               | Intrinsic PEEP            | mbar     | XX.X   | x |                 |                 |
| 7DH               | Peak Airway Pressure      | mbar     | *XXX_  | x |                 | x               |
| 81H               | Trapped Volume            | mL       | _XXX   | x |                 |                 |
| 82H <sup>3)</sup> | Expiratory Tidal Volume   | L        | X.XX   | x |                 | x <sup>2)</sup> |
| 88H <sup>3)</sup> | Expiratory Tidal Volume   | mL       | _XXX   | x |                 |                 |
| B5H               | Spont Breathing Frequency | 1/min    | XXX_   | x |                 |                 |
| B7H <sup>4)</sup> | Spont Minute Volume       | L/min    | XX.X   | x |                 |                 |
| 7AH <sup>5)</sup> | Spont Minute Volume       | L/min    | X.XX   | x |                 |                 |
| B8H <sup>6)</sup> | Minute Volume             | L/min    | X.XX   | x | x <sup>8)</sup> | x               |
| B9H <sup>7)</sup> | Minute Volume             | L/min    | XX.X   | x | x <sup>8)</sup> | x               |

1) Values below 100 will be transmitted with one decimal place, otherwise without a decimal place.

2) Device SW version 04.00 and higher:  
Limit is not transmitted, if alarm is switched off.

3) Values below 1 L will be transmitted in units of mL, otherwise in units of L.

4) Device SW version 01.XX.: Data is valid if measured value minute volume is valid.  
Device SW version 02.00 and higher: Data is valid if measured value minute volume is valid and exceeds or is equal to 1 L/min.

5) Available since Device SW version 02.00. Data is valid if measured value minute volume is valid and is lower than 1 L/min.

6) Available since Device SW version 02.00. Data is valid if measured value minute volume is valid and is lower than 1 L/min.  
Limits are valid if the value of the limit is lower than 1 L/min.

7) Device SW version 01.XX. Data is valid if measured value minute volume is valid. Limits are always valid.  
Device SW version 02.00 and higher: Data is valid if measured value minute volume is valid and exceeds or is equal to 1 L/min.  
Limits are valid if the value of the limit exceeds or is equal to 1 L/min.

8) Device SW version 04.00 and higher:  
Lower limit is not transmitted, if alarm is switched off.

A '\*' ahead the format indicates that the value may be negative. In that case a '-' character precedes the numeric value.

# MEDIBUS Specification for Evita 4 / Evita 2 dura

| Code                | Data-Description                                        | Unit      | Format | M | LL | HL |
|---------------------|---------------------------------------------------------|-----------|--------|---|----|----|
| C1H                 | Gas Temperature                                         | °C        | _XX_   | x |    |    |
| D6H                 | Breathing Frequency                                     | 1/min     | XXX_   | x |    | x  |
| 8BH                 | Inspiratory spont. Support-volume (VTASB) <sup>1)</sup> | mL        | XXXX   | x |    |    |
| 8DH <sup>1)2)</sup> | Negative Inspiratory Force                              | mbar      | *XXX   | x |    |    |
| C9H <sup>1)</sup>   | Rapid Shallow Breathing Index                           | 1/L x min | XXXX   | x |    |    |

1) Available since Device SW version 04.00

2) Device SW version 04.00 and higher:  
The value is not transmitted, if it is displayed on Evita 4 / Evita 2 dura as "< -45 mbar"

A '\*' ahead the format indicates that the value may be negative. In that case a '-' character precedes the numeric value.

## CO<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description            | Unit   | Format | M | LL | HL |
|------|-----------------------------|--------|--------|---|----|----|
| 09H  | CO <sub>2</sub> -Production | mL/min | XXX_   | x |    |    |
| 89H  | Dead Space                  | mL     | _XXX   | x |    |    |
| 8AH  | Relative Dead Space         | %      | _XX_   | x |    |    |
| DBH  | Endtidal CO <sub>2</sub>    | %      | XX.X   | x | x  | x  |
| E3H  | Endtidal CO <sub>2</sub>    | kPa    | XX.X   | x | x  | x  |
| E6H  | Endtidal CO <sub>2</sub>    | mmHg   | XXX_   | x | x  | x  |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

## O<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description     | Unit | Format | M | LL | HL |
|------|----------------------|------|--------|---|----|----|
| F0H  | Insp. O <sub>2</sub> | %    | XXX_   | x |    |    |

## SPO<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description | Unit  | Format | M | LL | HL |
|------|------------------|-------|--------|---|----|----|
| E1H  | Pulserate        | 1/min | XXX_   | x | x  | x  |
| EBH  | Saturation       | %     | XXX_   | x | x  | x  |

## Realtime Data

| Code | Realtime-Data                | Unit  |
|------|------------------------------|-------|
| 00H  | Airway Pressure              | mbar  |
| 01H  | Flow (insp./exp.)            | L/min |
| 02H  | Pleth                        | %     |
| 03H  | Resp. Vol. since insp. begin | mL    |
| 06H  | Exp. CO <sub>2</sub>         | mmHg  |
| 07H  | Exp. CO <sub>2</sub>         | kPa   |
| 08H  | Exp. CO <sub>2</sub>         | Vol%  |
| 20H  | Insp. Flow                   | L/min |
| 21H  | Exp. Flow                    | L/min |
| 24H  | Exp. Volume                  | mL    |

## Realtime Sync-Commands

Realtime Sync-Commands will be sent when Realtime Data are enabled.

| Command-Code | Argument | Command-Specification                 |
|--------------|----------|---------------------------------------|
| C6H          | C0H      | Start of Ventilator Inspiratory Cycle |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

## Alarm Messages

Alarm phrases are sent in english only.

Note: Some alarm priorities are configurable by user!

### Airway related Alarms

Codepage 1

| Code | Alarm Description                        | Priority | Alarm phrase  |
|------|------------------------------------------|----------|---------------|
| 10H  | Airway Pressure > high                   | 30       | PAW HIGH      |
| 19H  | Minute Volume < low Limit                | 29       | MIN VOL LOW   |
| 33H  | Volume not constant                      | 22       | VOL INCONST   |
| 3AH  | Assisted Spontaneous Breathing > 1.5 sec | 7        | ASB > 1,5 SEC |
| 3AH  | Assisted Spontaneous Breathing > 4 sec   | 28       | ASB > 4 SEC   |
| 42H  | Check Flow Sensor                        | 30       | FLOW SENSOR?  |
| 90H  | Respiratory Rate > high Limit            | 29       | RESP RATE HI  |
| 98H  | Apnoe detected by Evita                  | 29       | APNEA EVITA   |
| 9AH  | Disconnection Ventilator                 | 30       | PAW LOW       |
| 9BH  | Minute Volume > high Limit               | 29       | MIN VOL HIGH  |
| A4H  | Volume Measurement inoperable            | 31       | VOL ERR       |
| ACH  | MIN VOL Alarm disabled                   | 5        | MV ALRM OFF   |
| ADH  | Pressure Measurement inoperable          | 29       | PRESS ERR     |
| B8H  | Airway Temperature Measurement inop      | 27       | AW-TEMP INOP  |
| BAH  | Airway Temperature > high Limit          | 27       | AW-TEMP HI    |
| C4H  | Pressure limited Respiratory Volume      | 4        | PRESSURE LTD  |
| DAH  | PEEP > high Limit                        | 30       | PEEP HIGH     |
| E8H  | Tidal Volume > high Limit                | 8/29     | TIDVOL HI     |
| EAH  | Neo. Volume Measurement inoperable       | 1/25/31  | N-VOL ERR     |

Codepage 2

| Code | Alarm Description                   | Priority | Alarm phrase |
|------|-------------------------------------|----------|--------------|
| 6AH  | Tube obstruction                    | 30       | TUBE OBSTRUC |
| 66H  | Apnoe Alarm off                     | 5        | APN ALRM OFF |
| 67H  | Minute Volume Alarm low off         | 4        | MV LOW OFF   |
| 68H  | Tidal Volume Alarm high off         | 5        | VT HIGH OFF  |
| 6BH  | External Flow                       | 1        | EXTERN FLOW  |
| 90H  | Neonatal Flow Sensor not at Y-piece | 29/1     | NEO FLOW ?   |
| 93H  | Apnea Ventilation                   | 23       | APNEAE VENT  |
| 97H  | Check Slave Frequency Setting       | 2        | SLAVE FREQ ? |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

| Code | Alarm Description                                                                                                | Priority | Alarm phrase  |
|------|------------------------------------------------------------------------------------------------------------------|----------|---------------|
| 9AH  | Inspiratory Time in CPAP/PPS Mode > 1,5 sec                                                                      | 7        | PPS-TI > 1,5S |
| 9BH  | Inspiratory Time in ASB > T <sub>insp</sub> Setting<br>(neonatal mode only)                                      | 7        | ASB > TINSP   |
| 9CH  | Hose System Leaks                                                                                                | 1        | LEAKAGE       |
| 9DH  | Safety mode if neonatal flowsensor not available<br>during volume controlled ventilation<br>(neonatal mode only) | 30       | BACKUP VENT   |

## CO<sub>2</sub> related Alarms

Codepage 1

| Code | Alarm Description                              | Priority | Alarm phrase |
|------|------------------------------------------------|----------|--------------|
| 27H  | Endtidal CO <sub>2</sub> < low Limit           | 28       | ET CO2 LOW   |
| 28H  | Endtidal CO <sub>2</sub> > high Limit          | 28       | ET CO2 HIGH  |
| 3EH  | CO <sub>2</sub> Zero-Calibration requested     | 28       | CO2 ZERO CAL |
| 57H  | CO <sub>2</sub> Alarm disabled                 | 4        | CO2 ALRM OFF |
| 64H  | Clean CO <sub>2</sub> Sensor (Window occluded) | 28       | CLEAN CO2    |
| 6AH  | CO <sub>2</sub> Device Failure                 | 28       | CO2 ERR      |
| D9H  | CO <sub>2</sub> Sensor disconnected or fault   | 28       | CO2 SENSOR ? |

## O<sub>2</sub> related Alarms

Codepage 1

| Code | Alarm Description                           | Priority | Alarm phrase |
|------|---------------------------------------------|----------|--------------|
| 08H  | Insp. Oxygen < low Limit                    | 28       | % O2 LOW     |
| 37H  | % Oxygen > high Limit                       | 28       | % O2 HIGH    |
| A1H  | Insp. O <sub>2</sub> Measurement inoperable | 28       | % O2 ERR     |
| E9H  | O <sub>2</sub> Monitoring disabled          | 4        | O2 ALRM OFF  |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

## SpO<sub>2</sub> related alarms

Codepage 1

| Code | Alarm Description                             | Priority | Alarm phrase |
|------|-----------------------------------------------|----------|--------------|
| 01H  | No SpO <sub>2</sub> Pulse                     | 28       | NO SPO2 PULS |
| 02H  | SpO <sub>2</sub> Pulse < low Limit            | 6/28     | SPO2 PULS LO |
| 07H  | Oxygen Saturation < low Limit                 | 6/28     | SPO2 LOW     |
| 1EH  | SpO <sub>2</sub> Pulse > high Limit           | 6/28     | SPO2 PULS HI |
| 22H  | Oxygen Saturation > high Limit                | 6/28     | SPO2 HIGH    |
| 35H  | SpO <sub>2</sub> Sensor disconnected or fault | 28       | SPO2SEN DISC |
| 5BH  | Oximeter Alarm disable                        | 4        | SPO2 ALRM OF |
| 68H  | Oximeter Device Failure                       | 28       | SPO2 ERR     |

## Ventilator related Alarms

Codepage 1

| Code | Alarm Description                      | Priority | Alarm phrase |
|------|----------------------------------------|----------|--------------|
| 12H  | Check Air Supply                       | 10/31    | AIR SUPPLY ? |
| 13H  | O <sub>2</sub> Supply Press low        | 31       | LO O2 SUPPLY |
| 4BH  | Battery low or malfunction             | 6/31     | BATTERY ERR  |
| 9FH  | Problems with Respirator (Evita)       | 31       | EVITA ERR    |
| B0H  | Check Expiration-Valve                 | 30       | EXP-VALVE ?  |
| BDH  | Check O <sub>2</sub> Supply (Advisory) | 10       | LO O2 SUPPLY |
| C6H  | Respirator Synchronisation inoperable  | 27       | SYNCHRO INOP |
| C7H  | Fail to Cycle                          | 29       | CYCLE FAILED |
| C8H  | Gas Mixer inoperable (Alarm)           | 31       | MIXER INOP   |
| CAH  | Problems with Fan                      | 2/26     | FAN ERR      |
| E6H  | Air Supply Pressure > high Limit       | 3/16     | AIR PRESS HI |
| E7H  | High O <sub>2</sub> Supply Presssure   | 3/16     | HI O2 SUPPLY |

Codepage 2

| Code | Alarm Description             | Priority | Alarm phrase |
|------|-------------------------------|----------|--------------|
| 5AH  | Power Supply by Battery       | 10       | BATTERY ON   |
| 5CH  | Batterie less than 2 min left | 24       | BATT. < 2MIN |
| 91H  | Loss of Data                  | 30       | LOSS OF DATA |
| 94H  | Check Device                  | 21/22    | CHECK EVITA  |
| 95H  | Evita Standby                 | 31       | EVITA STDBY  |

| Code | Alarm Description            | Priority | Alarm phrase |
|------|------------------------------|----------|--------------|
| 3BH  | Ambient Pressure?            | 21       | AMB PRESS ?  |
| 3CH  | Error multi functional board | 1/21     | ERR MULTIPCB |
| 69H  | Remote Pad Error             | 1        | REM.PAD-ERR  |
| 96H  | PEEP Valve inoperable        | 30       | PEEP V ERR   |
| 98H  | Nebulizer active             | 3        | NEBULIZER ON |
| 99H  | Inspiration hold aborted     | 9        | INSPHOLD END |
| 9EH  | Expiration hold aborted      | 9        | EXSPHOLD END |
| 9FH  | Nebulizing aborted           | 17       | NEBULIZ OFF  |

## Device Settings

| Code | Setting Description      | Unit        | Format |
|------|--------------------------|-------------|--------|
| 01H  | Insp. O <sub>2</sub>     | %           | _XXX_  |
| 02H  | Max. insp Flow           | L/min       | XXX.X  |
| 04H  | Insp. Tidal Volume       | L           | X.XXX  |
| 05H  | Ti                       | sec         | XX.XX  |
| 07H  | I : E Insp               | -           | XXX.X  |
| 08H  | I : E Exp                | -           | XXX.X  |
| 09H  | Frequency                | 1/min       | XXX.X  |
| 0BH  | PEEP                     | mbar        | _XX.X  |
| 0CH  | Interm. PEEP             | mbar        | _XX_   |
| 0DH  | APRV Low Pressure        | mbar        | _XX_   |
| 0EH  | APRV High Pressure       | mbar        | _XX_   |
| 0FH  | APRV Low Time            | sec         | _XX.X  |
| 10H  | APRV High Time           | sec         | _XX.X  |
| 11H  | Apnea Time               | sec         | _XX_   |
| 12H  | ASB                      | mbar        | _XX.X  |
| 13H  | Max./Insp. Pressure      | mbar        | XXX.X  |
| 16H  | Tachyapnea Frequency     | 1/min       | _XXX_  |
| 29H  | Flow Trigger             | L/min       | _XX.X  |
| 2EH  | ASB Ramp                 | sec         | _X.XX  |
| 3CH  | Flow Assist              | mbar*s/L/10 | XXXXX  |
| 3DH  | Volume Assist            | mbar/L/10   | XXXXX  |
| 4EH  | TDeconnect <sup>1)</sup> | sec         | _XX_   |

1) Available since Device SW version 04.00.

## Text Messages

Text Messages are sent in english only.

| Code              | Text Message                 |
|-------------------|------------------------------|
| 01H               | Mode IPPV                    |
| 02H               | Mode IPPV/ASSIST             |
| 06H               | Mode SIMV                    |
| 07H               | Mode SIMV/ASB                |
| 0EH               | Mode BIPAP                   |
| 2DH               | Mode BIPAP/ASB               |
| 2EH               | Mode SIMV/AutoFlow           |
| 2FH               | Mode SIMV/ASB/AutoFlow       |
| 30H               | Mode IPPV/AutoFlow           |
| 31H               | Mode IPPV/ASSIST/AutoFlow    |
| 1AH               | Mode APRV                    |
| 0CH               | Mode MMV                     |
| 0DH               | Mode MMV/ASB                 |
| 32H               | Mode MMV/AutoFlow            |
| 33H               | Mode MMV/ASB/AutoFlow        |
| 0AH               | Mode CPAP                    |
| 0BH               | Mode CPAP/ASB                |
| 11H               | Mode APNEA VENTILATION       |
| 35H               | Mode CPAP/PPS                |
| 0FH               | Mode SYNCHRON MASTER         |
| 10H               | Mode SYNCHRON SLAVE          |
| 47H <sup>2)</sup> | Mode BIPAP/ASSIST            |
| 20H               | Mode Adults                  |
| 3AH               | Mode Pediatrics              |
| 21H               | Mode Neonates                |
| 48H <sup>2)</sup> | IV-Invasive Ventilation      |
| 49H <sup>2)</sup> | NIV-Non-Invasive Ventilation |
| 22H <sup>1)</sup> | mmHg                         |
| 23H <sup>1)</sup> | kPa                          |
| 24H <sup>1)</sup> | %                            |

1) Selected CO<sub>2</sub> unit

2) Available since Device SW version 04.00

## Device specific Commands

Evita 4 uses the device specific command (code 6AH) to transmit some additional information which is not part of the MEDIBUS standard. The device specific command has the following format:

| ESC    | Command Code (6AH) | Argument | Checksum | CR |
|--------|--------------------|----------|----------|----|
| 0      | 1                  | 2        | 3        | 5  |
| 6 Byte |                    |          |          |    |

The 'Argument' is a one byte ASCII character specifying the device specific command:

| Argument | Meaning                    |
|----------|----------------------------|
| '2'      | Request Key Status         |
| '3'      | Request SW Version         |
| '4'      | Request Patient Type       |
| '5'      | Request Evita Link Channel |

## Device specific Responses

### Key Status

This response returns all active keys on a 'Request Key Status' command in the following format:

| SOH       | Command Echo (6AH) | Argument Echo ('2') | Code 1 | ... | Code m | Checksum | CR   |
|-----------|--------------------|---------------------|--------|-----|--------|----------|------|
| 0         | 1                  | 2                   | 3      | 5   | 2m+1   | 2m+3     | 2m+4 |
| 2m+5 Byte |                    |                     |        |     |        |          |      |

The Code is a two byte ASCII field holding an identifier for each currently active key. The meaning of the Code is given in the following table:

| Key Code | Meaning                               |
|----------|---------------------------------------|
| '32'     | Alarm Silence active                  |
| '33'     | Nebulizer active                      |
| '35'     | Oxygen Calibration active             |
| '37'     | Suction active                        |
| '38'     | Flow Calibration active               |
| '39'     | CO2 Calibration, Zero or Check active |
| '40'     | Monitoring Flow active                |
| '41'     | Monitoring FiO2 active                |
| '42'     | Monitoring SpO2 active                |
| '43'     | Monitoring CO2 active                 |
| '44'     | Monitoring Paed. Flow active          |

# MEDIBUS Specification for Evita 4 / Evita 2 dura

## EvitaLink Version

This response returns the software version of EvitaLink in the following format:

| SOH     | Command Echo (6AH) | Argument Echo ('3') | Version | Checksum | CR |
|---------|--------------------|---------------------|---------|----------|----|
| 0       | 1                  | 2                   | 3       | 8        | 9  |
| 10 Byte |                    |                     |         |          |    |

The Version is a five byte ASCII field.  
Example: Version 2.00 = '02.00'

## Patient Type

This response returns the type of patient which has been chosen in Evita 4 in the following format:

| SOH      | Command Echo (6AH) | Argument Echo ('4') | Patient | Checksum | CR  |
|----------|--------------------|---------------------|---------|----------|-----|
| 0        | 1                  | 2                   | 3       | m+3      | m+4 |
| m+5 Byte |                    |                     |         |          |     |

Patient is a ASCII byte field which is given in the table below:

| Patient    | ASCII Byte field | Field lenght |
|------------|------------------|--------------|
| Adult      | 'Adult'          | 5            |
| Paediatric | 'Paediatric'     | 10           |
| Neonates   | 'Neonates'       | 8            |

## EvitaLink Channel

This response returns the number of the EvitaLink channel as a one byte ASCII Code:

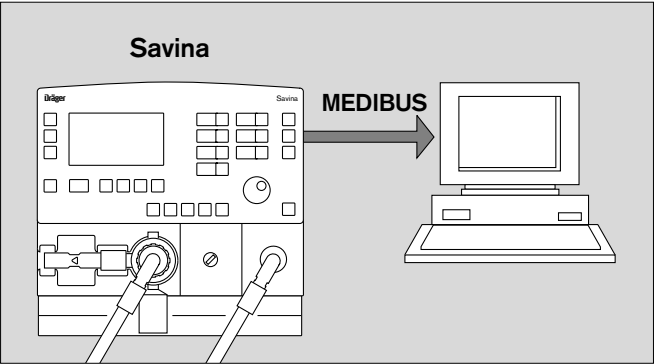
| Channel | ASCII Code |
|---------|------------|
| Com1    | '0'        |
| Com2    | '1'        |
| Com3    | '2'        |

The response format is specified as follows:

| SOH    | Command Echo (6AH) | Argument Echo ('5') | Channel | Checksum | CR |
|--------|--------------------|---------------------|---------|----------|----|
| 0      | 1                  | 2                   | 3       | 4        | 5  |
| 6 Byte |                    |                     |         |          |    |

# MEDIBUS Specification for Savina

## Device Connection



## Port Specification

|                    |                        |
|--------------------|------------------------|
| Connector          |                        |
| Type               | RS-232-C               |
| Pins               | 9 pin Sub D (female)   |
| Pins               | 2 RXD                  |
| Pins               | 3 TXD                  |
| Pins               | 5 GND                  |
| Galvanic Isolation | 1,5 kV                 |
| Location           | rear side of Savina    |
| Label RS 232       |                        |
| Port Configuration |                        |
| Baudrate           | 1200, 2400, 4800, 9600 |
| Baudrate           | 19200 Baud             |
| Databits           | 8                      |
| Startbits          | 1                      |
| Stopbits           | 1, 2                   |
| Parity             | no, odd, even          |

## Device Identification for Savina

|                 |                                           |
|-----------------|-------------------------------------------|
| ID-Number       | 8250                                      |
| Name            | 'Savina'                                  |
| MEDIBUS-Version | 04.00 for Device Version 01.00 and higher |

# MEDIBUS Specification for Savina

## Available Data

MEDIBUS is available since Savina device version 01.00 and higher.

All messages are transfered in english only independent of the language setting.

## Commands

### Transmitted Commands

| Code | Command-Specification          |
|------|--------------------------------|
| 30H  | Do nothing (NOP)               |
| 49H  | Time changed                   |
| 51H  | Initialize Communication (ICC) |
| 52H  | Request Device Identification  |

### Processed and responded Command

| Code | Command-Specification               |
|------|-------------------------------------|
| 24H  | Request current DATA                |
| 25H  | Request current LOW ALARM LIMITS    |
| 26H  | Request current HIGH ALARM LIMITS   |
| 27H  | Request current ALARMS (Codepage 1) |
| 28H  | Request Current Date And Time       |
| 29H  | Request current DEVICE SETTINGS     |
| 2AH  | Request current TEXT MESSAGES       |
| 2EH  | Request current ALARMS (Codepage 2) |
| 30H  | Do nothing (NOP)                    |
| 4AH  | Configure Data Response             |
| 51H  | Initialize Communication (ICC)      |
| 52H  | Request Device Identification       |
| 53H  | Request Realtime Configuration      |
| 54H  | Configure Realtime Transmission     |
| 55H  | Stop Communication                  |
| 6AH  | Device Specific                     |

## Measured Data, Low and High Alarm Limits

Airway related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code              | Data-Description             | Unit     | Format | M | LL              | HL              |
|-------------------|------------------------------|----------|--------|---|-----------------|-----------------|
| 07H <sup>1)</sup> | Compliance                   | L/bar    | _XXX   | x |                 |                 |
| 08H <sup>1)</sup> | Resistance                   | mbar/L/s | XXX_   | x |                 |                 |
| 69H               | Plateau (Tplat)              | s        | XX.X   | x |                 |                 |
| 6FH               | Inspiratory Time             | s        | XX.X   | x |                 |                 |
| 73H               | Mean Breathing Pressure      | mbar     | *_XX_  | x |                 |                 |
| 74H               | Plateau Pressure             | mbar     | *_XX_  | x |                 |                 |
| 76H               | Flow Peak                    | mL/s     | XXXX   | x |                 |                 |
| 78H               | PEEP Breathing Pressure      | mbar     | *_XX_  | x |                 |                 |
| 7AH <sup>2)</sup> | Spontaneous Minute Volume    | L/min    | X.XX   | x |                 |                 |
| B7H <sup>3)</sup> | Spontaneous Minute Volume    | L/min    | XX.X   | x |                 |                 |
| 7DH               | Peak Breathing Pressure      | mbar     | *XXX_  | x |                 | x               |
| 88H               | Tidal Volume in mL           | mL       | XXXX   | x |                 | x <sup>4)</sup> |
| B5H               | Spontaneous Respiratory Rate | 1/min    | XXX_   | x |                 |                 |
| B8H <sup>2)</sup> | Respiratory Minute Volume    | L/min    | X.XX   | x |                 |                 |
| B9H <sup>3)</sup> | Respiratory Minute Volume    | L/min    | XX.X   | x | x <sup>4)</sup> | x               |
| C1H               | Airway Temperatur            | °C       | _XX_   | x |                 |                 |
| D6H               | Respiratory Rate             | 1/min    | XXX_   | x |                 | x               |
| E7H               | I:E I-Part                   | –        | XX.X   | x |                 |                 |
| E8H               | I:E E-Part                   | –        | XX.X   | x |                 |                 |

1) Available since Device SW version 02.00.

2) Measured Data is valid if measured value minute volume is valid and is lower than 1 L/min.

3) Measured Data is valid if measured value minute volume is valid and exceeds or is equal to 1 L/min.

4) Device SW version 02.00 and higher: Limit is not transmitted, if alarm is switched off.

# MEDIBUS Specification for Savina

## O<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description     | Unit | Format | M | LL | HL |
|------|----------------------|------|--------|---|----|----|
| F0H  | Insp. O <sub>2</sub> | %    | XXX_   | x | x  | x  |

## Realtime Data

| Code | Realtime-Data                | Unit  |
|------|------------------------------|-------|
| 00H  | Airway Pressure              | mbar  |
| 01H  | Flow                         | L/min |
| 03H  | Resp. Vol. since insp. begin | mL    |

## Realtime Sync-Commands

Realtime Sync-Commands will be sent when Realtime Data are enabled.

| Command-Code | Argument | Command-Specification                 |
|--------------|----------|---------------------------------------|
| C6H          | C0H      | Start of Ventilator Inspiratory Cycle |

## Alarm Messages

Alarm phrases are sent in english only.

Note: Some alarm priorities are configurable by user!

### Airway related Alarms

Codepage 1

| Code | Alarm Description                         | Priority | Alarm phrase |
|------|-------------------------------------------|----------|--------------|
| 10H  | Airway Pressure > high                    | 30       | PAW HIGH     |
| 19H  | Minute Volume < low Limit                 | 29       | MIN VOL LOW  |
| 33H  | Tidal Volume < VT <sub>insp</sub> setting | 8/22     | TIDVOL LO    |
| 3AH  | Assisted Spontaneous Breathing > 4 sec    | 5/28     | ASB > 4 SEC  |
| 42H  | Check Flow Sensor                         | 30       | FLOW SENSOR? |
| 90H  | Respiratory Rate > high Limit             | 29       | RESP RATE HI |
| 98H  | Apnoe detected by Savina                  | 29       | APNEA SAVINA |
| 9AH  | Disconnection Ventilator                  | 30       | PAW LOW      |
| 9BH  | Minute Volume > high Limit                | 29       | MIN VOL HIGH |
| A4H  | Volume Measurement inoperable             | 30       | VOL ERR      |
| ACH  | MIN VOL Alarm disabled                    | 5        | MV ALARM OFF |
| ADH  | Pressure Measurement inoperable           | 30       | PRESS ERR    |
| B8H  | Airway Temperature Measurement inop       | 27       | AW-TEMP INOP |
| BAH  | Airway Temperature > high Limit           | 7/27     | AW-TEMP HI   |
| DAH  | PEEP > high Limit                         | 30       | PEEP HIGH    |
| E8H  | Tidal Volume > high Limit                 | 8/29     | TIDVOL HI    |

Codepage 2

| Code              | Alarm Description           | Priority | Alarm phrase |
|-------------------|-----------------------------|----------|--------------|
| 66H <sup>1)</sup> | Apnoe Alarm off             | 5        | APN ALRM OFF |
| 67H <sup>1)</sup> | Minute Volume Alarm low off | 4        | MV LOW OFF   |
| 68H <sup>1)</sup> | Tidal Volume Alarm high off | 5        | VT HIGH OFF  |

1) Available since Device SW version 02.00.

# MEDIBUS Specification for Savina

## O<sub>2</sub> related Alarms

Codepage 1

| Code | Alarm Description                           | Priority | Alarm phrase |
|------|---------------------------------------------|----------|--------------|
| 08H  | Insp. Oxygen < low Limit                    | 28       | % O2 LOW     |
| 37H  | Insp. Oxygen > high Limit                   | 28       | % O2 HIGH    |
| A1H  | Insp. O <sub>2</sub> Measurement inoperable | 28       | % O2 ERR     |
| E9H  | O <sub>2</sub> Monitoring disabled          | 4/22     | O2 ALARM OFF |

## Ventilator related Alarms

Codepage 1

| Code | Alarm Description                    | Priority | Alarm phrase |
|------|--------------------------------------|----------|--------------|
| 13H  | O <sub>2</sub> Supply Press low      | 6/30     | LO O2 SUPPLY |
| 4BH  | Battery low or malfunction           | 6/31     | BATTERY ERR  |
| 9FH  | Problems with Respirator (Savina)    | 31       | SAVINA ERR   |
| B0H  | Check Expiration-Valve               | 30       | EXP-VALVE ?  |
| C7H  | Fail to Cycle                        | 29       | CYCLE FAILED |
| C9H  | Check Cooling                        | 21       | COOLING ?    |
| CAH  | Problems with Fan                    | 27       | FAN ERR      |
| E7H  | High O <sub>2</sub> Supply Presssure | 3/16     | HI O2 SUPPLY |

Codepage 2

| Code | Alarm Description                | Priority | Alarm phrase |
|------|----------------------------------|----------|--------------|
| 5AH  | Power Supply by Internal Battery | 6/24     | INT BATT ON  |
| 5CH  | Battery low                      | 6/21/31  | BATTERY LOW  |
| 6CH  | Check Microfilter                | 2/20/30  | MICROFILTER? |
| 6DH  | Power Supply by External DC      | 2        | EXTERN DC ON |
| 6EH  | Ambient Pressure high            | 6        | AMB PRESS HI |
| 6FH  | Ambient Pressure low             | 6        | AMB PRESS LO |
| 93H  | Apnea Ventilation                | 23       | APNEAE VENT  |
| 94H  | Check Savina                     | 21       | CHECK SAVINA |
| 95H  | Savina Standby                   | 31       | SAVINA STDBY |
| 96H  | Problems with PEEP control       | 29       | PEEP ERR     |
| 98H  | Nebulizer active                 | 3        | NEBULIZER ON |
| 99H  | Inspiration hold aborted         | 9        | INSPHOLD END |
| 9CH  | Leakage                          | 1        | LEAKAGE      |

## Device Settings

| Code | Setting Description                 | Unit   | Format |
|------|-------------------------------------|--------|--------|
| 01H  | Insp. O <sub>2</sub>                | %      | _XXX_  |
| 04H  | Insp. Tidal Volume                  | L      | X.XXX  |
| 05H  | Ti                                  | s      | XX.XX  |
| 07H  | I : E Insp                          | -      | XXX.X  |
| 08H  | I : E Exp                           | -      | XXX.X  |
| 09H  | Frequency                           | 1/min  | XXX.X  |
| 0BH  | PEEP                                | mbar   | _XX.X  |
| 0CH  | Interm. PEEP                        | mbar   | _XX.X  |
| 11H  | Apnea Time                          | sec    | __XX_  |
| 12H  | ASB                                 | mbar   | _XX.X  |
| 13H  | Max. insp. Airway Pressure (Pmax)   | mbar   | XXX.X  |
| 29H  | Flow Trigger                        | L/min  | _XX.X  |
| 42H  | Backup Frequency                    | 1/min  | XXX.X  |
| 44H  | Backup Tidal Volume                 | L      | X.XXX  |
| 45H  | Insp. Pressure (P <sub>insp</sub> ) | mbar   | XXX.X  |
| 4EH  | TDeconnect <sup>1)</sup>            | sec    | __XX_  |
| 4FH  | Flow Acceleration                   | mbar/s | __XXX  |

1) Available since Device SW version 02.00.

## Text Messages

Text Messages are sent in english only.

| Code              | Text Message                   |
|-------------------|--------------------------------|
| 01H               | Mode IPPV                      |
| 02H               | Mode IPPV/ASSIST               |
| 06H               | Mode SIMV                      |
| 07H               | Mode SIMV/ASB                  |
| 08H               | Mode Standby                   |
| 0AH               | Mode CPAP                      |
| 0BH               | Mode CPAP/ASB                  |
| 0EH               | Mode BIPAP                     |
| 11H               | Mode APNEA VENTILATION         |
| 2DH               | Mode BIPAP/ASB                 |
| 2EH               | Mode SIMV/AutoFlow             |
| 2FH               | Mode SIMV/ASB/AutoFlow         |
| 30H               | Mode IPPV/AutoFlow             |
| 31H               | Mode IPPV/ASSIST/AutoFlow      |
| 48H <sup>1)</sup> | IV – Invasive Ventilation      |
| 49H <sup>1)</sup> | NIV – Non-Invasive Ventilation |

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1) Available since Device SW version 02.00.

# MEDIBUS Specification for Savina

## Device specific Commands

Savina uses the device specific command (code 6AH) to transmit some additional information which is not part of the MEDIBUS standard. The device specific command has the following format:

| ESC    | Command Code (6AH) | Argument | Checksum | CR |
|--------|--------------------|----------|----------|----|
| 0      | 1                  | 2        | 3        | 5  |
| 6 Byte |                    |          |          |    |

The 'Argument' is a one byte ASCII character specifying the device specific command:

| Argument | Meaning                     |
|----------|-----------------------------|
| '2'      | Request Key Status          |
| '3'      | Request Savina Link Version |

## Device specific Responses

### Key Status

This response returns all active keys on a 'Request Key Status' command in the following format:

| SOH       | Command Echo (6AH) | Argument Echo ('2') | Code 1 | ... | Code m | Checksum | CR   |
|-----------|--------------------|---------------------|--------|-----|--------|----------|------|
| 0         | 1                  | 2                   | 3      | 5   | 2m+1   | 2m+3     | 2m+4 |
| 2m+5 Byte |                    |                     |        |     |        |          |      |

The Code is a two byte ASCII field holding an identifier for each currently active key. The meaning of the Code is given in the following table:

| Key Code | Meaning                   |
|----------|---------------------------|
| '32'     | Alarm Silence active      |
| '33'     | Nebulizer active          |
| '35'     | Oxygen Calibration active |
| '37'     | Suction active            |
| '40'     | Monitoring Flow active    |
| '41'     | Monitoring FiO2 active    |

# MEDIBUS Specification for Savina

## Savina Link Version

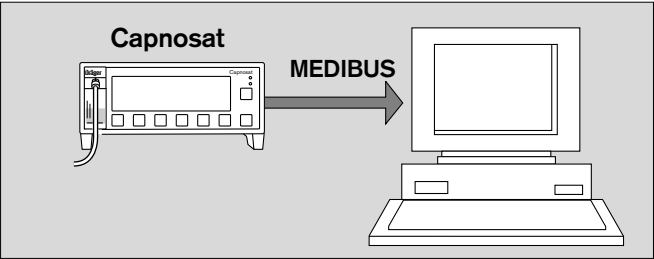
This response returns the software version of Savina Link in the following format:

| SOH     | Command Echo (6AH) | Argument Echo ('3') | Version | Checksum | CR |
|---------|--------------------|---------------------|---------|----------|----|
| 0       | 1                  | 2                   | 3       | 8        | 9  |
| 10 Byte |                    |                     |         |          |    |

The Version is a five byte ASCII field.  
Example: Version 1.00 = '01.00'

# MEDIBUS Specification for Capnosat

## Device Connection



## Port Specification

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| Connector          |                                                                                |
| Type               | RS-232-C                                                                       |
|                    | 25 pin Sub D (female)                                                          |
| Pins               | 1 Housing                                                                      |
|                    | 2 TXD                                                                          |
|                    | 3 RXD                                                                          |
|                    | 7 GND                                                                          |
| Galvanic Isolation | none: Only devices fulfilling IEC 601 are allowed to be connected to Capnosat. |
| Location           | rear side of Capnosat                                                          |
|                    | Label: RS-232-C                                                                |
| Port Configuration |                                                                                |
| Baudrate           | 9600 Baud                                                                      |
| Databits           | 8                                                                              |
| Startbits          | 1                                                                              |
| Stopbits           | 1                                                                              |
| Parity             | even                                                                           |

## Device Identification

|                 |                                           |
|-----------------|-------------------------------------------|
| ID-Number       | 8300                                      |
| Name            | 'Capnosat'                                |
| MEDIBUS-Version | 03.00 for Device Version 01.00 and higher |

## Available Data

Current Measured Data, Low and High Alarm Limits, Alarmstatus and Realtime Data are available since Capnosat device version 1.00.

# MEDIBUS Specification for Capnosat

## Commands

### Transmitted Commands

| Code | Command-Specification          |
|------|--------------------------------|
| 30H  | Do nothing (NOP)               |
| 51H  | Initialize Communication (ICC) |
| 52H  | Request Device Identification  |

### Processed and responded Commands

| Code | Command-Specification             |
|------|-----------------------------------|
| 24H  | Request current DATA              |
| 25H  | Request current LOW ALARM LIMITS  |
| 26H  | Request current HIGH ALARM LIMITS |
| 27H  | Request current ALARMS            |
| 30H  | Do nothing (NOP)                  |
| 4AH  | Configure Data Response           |
| 51H  | Initialize Communication (ICC)    |
| 52H  | Request Device Identification     |
| 53H  | Request Realtime Configuration    |
| 54H  | Configure Realtime Transmission   |
| 55H  | Stop Communication                |

## Measured Data, Low and High Alarm Limits

### CO<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description                    | Unit  | Format | M | LL | HL |
|------|-------------------------------------|-------|--------|---|----|----|
| D5H  | Respiratory Rate (CO <sub>2</sub> ) | 1/min | XX__   | x |    |    |
| DAH  | Insp. CO <sub>2</sub> in %          | %     | XX.X   | x |    | x  |
| DBH  | Endtidal CO <sub>2</sub> in %       | %     | XX.X   | x | x  | x  |
| E3H  | Endtidal CO <sub>2</sub> in kPa     | kPa   | XX.X   | x | x  | x  |
| E5H  | Insp. CO <sub>2</sub> mmHg          | mmHg  | XX__   | x |    | x  |
| E6H  | Endtidal CO <sub>2</sub> in mmHg    | mmHg  | XX__   | x | x  | x  |
| FFH  | Insp. CO <sub>2</sub> kPa           | kPa   | XX.X   | x |    | x  |

# MEDIBUS Specification for Capnosat

## SpO<sub>2</sub> related Measured Data (M), Low (LL) and High (HL) Alarm Limits

| Code | Data-Description      | Unit  | Format | M | LL | HL |
|------|-----------------------|-------|--------|---|----|----|
| E1H  | Pulse Rate (OXIMETER) | 1/min | XXX_   | x | x  | x  |
| EBH  | Oxygen Saturation     | %     | XXX_   | x | x  | x  |

## Realtime-Data

| Code | Realtime-Data                    | Unit         |
|------|----------------------------------|--------------|
| 02H  | Oxygen Saturation Pulse (Pleth.) | % Full Scale |
| 06H  | CO <sub>2</sub>                  | mmHg         |
| 07H  | CO <sub>2</sub>                  | kPa          |
| 08H  | CO <sub>2</sub>                  | %            |

## Alarm Messages

### CO<sub>2</sub> related Alarms:

|       |                            |                                                  |                          |  |
|-------|----------------------------|--------------------------------------------------|--------------------------|--|
|       | Priority: 31               | Apnea - No CO <sub>2</sub> fluct. for 30 Seconds |                          |  |
| Code: | D: APNOE CO <sub>2</sub>   | GB: APNEA CO <sub>2</sub>                        | F: APNEE CO <sub>2</sub> |  |
| 0DH   | I: APNEA - CO <sub>2</sub> | NL: APNOE CO <sub>2</sub>                        | E: APNEA CO <sub>2</sub> |  |

|       |                           |                                      |                           |  |
|-------|---------------------------|--------------------------------------|---------------------------|--|
|       | Priority: 18              | Endtidal CO <sub>2</sub> < low Limit |                           |  |
| Code: | D: ET CO <sub>2</sub> \$& | GB: ET CO <sub>2</sub> LOW           | F: ET CO <sub>2</sub> \$& |  |
| 27H   | I: ET CO <sub>2</sub> \$& | NL: ET CO <sub>2</sub> \$&           | E: CO <sub>2</sub> FE \$& |  |

|       |                          |                                       |                          |  |
|-------|--------------------------|---------------------------------------|--------------------------|--|
|       | Priority: 18             | Endtidal CO <sub>2</sub> > high Limit |                          |  |
| Code: | D: ET CO <sub>2</sub> "# | GB: ET CO <sub>2</sub> HIGH           | F: ET CO <sub>2</sub> "# |  |
| 28H   | I: ET CO <sub>2</sub> "# | NL: ET CO <sub>2</sub> "#             | E: CO <sub>2</sub> FE "# |  |

# MEDIBUS Specification for Capnosat

|       |                |                              |                |  |
|-------|----------------|------------------------------|----------------|--|
|       | Priority: 2    | Inspiratory CO2 > high Limit |                |  |
| Code: | D: INSP CO2 "# | GB: INSP CO2 HI              | F: CO2 INSP "# |  |
| 3CH   | I: CO2 INSP "# | NL: INSP CO2 "#              | E: CO2 INSP "# |  |

|       |                 |                                 |                |  |
|-------|-----------------|---------------------------------|----------------|--|
|       | Priority: 7     | CO2 Patient Sensor Line blocked |                |  |
| Code: | D: CO2 LEITUNG? | GB: CO2 LINE BLK                | F: TUYAU CO2 ? |  |
| 3DH   | I: LINEA CO2 ?  | NL: CO2 LEIDING?                | E: TOMA CO2 ?  |  |

|       |                 |                          |                 |  |
|-------|-----------------|--------------------------|-----------------|--|
|       | Priority: 8     | Apnea CO2 Alarm disabled |                 |  |
| Code: | D: APN.CO2'@AUS | GB: APN.CO2'@OFF         | F: ARR'@APN CO2 |  |
| 56H   | I: APN.CO2'@OFF | NL: APN.CO2'@UIT         | E: APNCO2'@DESC |  |

|       |               |                    |                 |  |
|-------|---------------|--------------------|-----------------|--|
|       | Priority: 1   | CO2 Alarm disabled |                 |  |
| Code: | D: CO2 '@ AUS | GB: CO2 ALRM OFF   | F: ARRET '@ CO2 |  |
| 57H   | I: '@ CO2 OFF | NL: CO2 '@ UIT     | E: '@ CO2 DESC  |  |

|       |                 |                                     |               |  |
|-------|-----------------|-------------------------------------|---------------|--|
|       | Priority: 1     | CO2 Mon. in Low Acc. Mode (warm up) |               |  |
| Code: | D: CO2 AUFHEIZ  | GB: CO2 WARM UP                     | F: PRECHF CO2 |  |
| 63H   | I: CO2 RISCALD. | NL: CO2 OPWARM                      | E: CALEN CO2  |  |

|       |             |                    |             |  |
|-------|-------------|--------------------|-------------|--|
|       | Priority: 1 | CO2 Device Failure |             |  |
| Code: | D: CO2 INOP | GB: CO2 ERR        | F: CO2 INOP |  |
| 6AH   | I: CO2 INOP | NL: CO2 INOP       | E: CO2 INOP |  |

## SpO2 related Alarms

|       |                  |                        |                  |  |
|-------|------------------|------------------------|------------------|--|
|       | Priority: 31     | SpO2 Pulse < low Limit |                  |  |
| Code: | D: PULS SPO2 \$& | GB: SPO2 PULS LO       | F: POULS SAT \$& |  |
| 02H   | I: POLSO SPO2\$& | NL: POLS SPO2 \$&      | E: PULSO SPO2\$& |  |

|       |              |                               |             |  |
|-------|--------------|-------------------------------|-------------|--|
|       | Priority: 31 | Oxygen Saturation < low Limit |             |  |
| Code: | D: SPO2 \$&  | GB: SPO2 LOW                  | F: SPO2 \$& |  |
| 07H   | I: SPO2 \$&  | NL: SPO2 \$&                  | E: SPO2 \$& |  |

# MEDIBUS Specification for Capnosat

|       |                 |                         |                 |  |
|-------|-----------------|-------------------------|-----------------|--|
| Code: | Priority: 21    | SpO2 Pulse > high Limit |                 |  |
|       | D: PULS SPO2 "# | GB: SPO2 PULS HI        | F: POULS SAT "# |  |
|       | I: POLSO SPO2"# | NL: POLS SPO2 "#        | E: PULSO SPO2"# |  |

|       |              |                                |            |  |
|-------|--------------|--------------------------------|------------|--|
| Code: | Priority: 21 | Oxygen Saturation > high Limit |            |  |
|       | D: SPO2 "#   | GB: SPO2 HIGH                  | F: SPO2 "# |  |
|       | I: SPO2 "#   | NL: SPO2 "#                    | E: SPO2 "# |  |

|       |                 |                                   |                 |  |
|-------|-----------------|-----------------------------------|-----------------|--|
| Code: | Priority: 10    | SpO2 Sensor disconnected or fault |                 |  |
|       | D: SPO2 SENSOR? | GB: SPO2SEN DISC                  | F: CAPT SPO2 ?  |  |
|       | I: SENSOR SPO2? | NL: SPO2 SENSOR?                  | E: SENSOR SPO2? |  |

|       |                |                         |                 |  |
|-------|----------------|-------------------------|-----------------|--|
| Code: | Priority: 1    | Oximeter Alarm disabled |                 |  |
|       | D: SPO2 '@ AUS | GB: SPO2 ALRM OF        | F: ARRET '@ SAT |  |
|       | I: '@ SPO2 OFF | NL: SPO2 '@ UIT         | E: '@ SPO2 DESC |  |

|       |              |                         |              |  |
|-------|--------------|-------------------------|--------------|--|
| Code: | Priority: 1  | Oximeter Device Failure |              |  |
|       | D: SPO2 INOP | GB: SPO2 ERR            | F: SPO2 INOP |  |
|       | I: SPO2 INOP | NL: SPO2 INOP           | E: SPO2 INOP |  |

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Subject to alteration